

Lab 40: Capstone — Complete Machine Learning Pipeline with Visualisation

Aim

To integrate **NumPy, Pandas, a machine learning model, and visualisation** into one complete end-to-end machine learning pipeline using a dataset.

Algorithm

1. Load the dataset using Pandas.
2. Explore and display the dataset summary.
3. Prepare the features and target values.
4. Split the dataset into training and testing sets.
5. Train two machine learning models: **Decision Tree and KNN**.
6. Evaluate both models using accuracy.
7. Compare the model accuracies.
8. Visualise the results using a bar chart.

Program

Lab 40: Complete Machine Learning Pipeline with Visualisation

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.datasets import load_iris
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.tree import DecisionTreeClassifier
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.metrics import accuracy_score
```

```
# Load the Iris dataset

iris = load_iris()

# Create DataFrame

df = pd.DataFrame(
    iris.data,
    columns=iris.feature_names
)

df["target"] = iris.target

# Display dataset summary

print("Dataset Summary:\n")
print(df.describe())

# Features and target

X = df.iloc[:, :-1]
y = df["target"]

# Split the dataset

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=1
)

# Create models

models = {
    "Decision Tree": DecisionTreeClassifier(random_state=1),
    "KNN": KNeighborsClassifier(n_neighbors=3)
```

```
}
```

```
# Store accuracies
```

```
accuracies = {}
```

```
# Train and evaluate models
```

```
for name, model in models.items():
```

```
    model.fit(X_train, y_train)
```

```
    predictions = model.predict(X_test)
```

```
    accuracy = accuracy_score(y_test, predictions)
```

```
    accuracies[name] = accuracy
```

```
    print(f"\n{name} Accuracy: {accuracy}")
```

```
# Visualise model comparison
```

```
plt.bar(
```

```
    accuracies.keys(),
```

```
    accuracies.values()
```

```
)
```

```
plt.title("Model Accuracy Comparison")
```

```
plt.xlabel("Models")
```

```
plt.ylabel("Accuracy")
```

```
plt.ylim(0, 1.1)
```

```
plt.show()
```

Sample Output

Dataset Summary:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333	1.000000
...					

Decision Tree Accuracy: 0.9555555555555556

KNN Accuracy: 0.9777777777777777

A **bar chart comparing the accuracy of the Decision Tree and KNN models** will also be displayed.

Result

Thus, a **complete end-to-end machine learning pipeline was successfully implemented**, including dataset loading, exploration, model training, evaluation, comparison, and visualisation.